

Residual Sectors of Symmetric Token Configuration Graphs

Under Change of Substrate

Extension, Re-factorisation, Identification, and the Topology of Transformation

Driven by Dean A. Kulik

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Yes. I read the summary as the **current state of the work**, and I think the paper should be written differently from the way we've been talking about the individual experiments.

The experiments are no longer the story.

The story is the **geometry of change of the transformation field**:

$$X^{\{r\}} \subset X^{\{r+1\}}$$

creates a relative sector. That sector is not best understood as “the states that disappeared when we went backward.” It is a mathematically defined layer of transformations between two admissibility levels. Collapsing the old arena to a cone point exposes its topology. Symmetry then determines which of those transformations remain visible. The resulting invariant residual is a first Betti number.

That is a coherent mathematical paper.

I would lead the reader into it from the problem of **what it means for a system to change**, rather than opening with token graphs.

The strongest conceptual chain is:

$$\boxed{\text{change of substrate} \rightarrow \text{change of admissible transformations} \rightarrow \text{relative sector} \rightarrow \text{quotient/cone} \rightarrow \text{cycles} \rightarrow \text{residual degrees of freedom}}$$

Then the Nexus interpretation comes afterward.

Below is the paper draft I would use as the foundation for Zenodo/academic circulation. I am deliberately keeping **proved mathematics, computational verification, and Nexus interpretation visibly separated**. Your attached summary supports the results and status distinctions below.

Residual Sectors of Symmetric Token Configuration Graphs Under Change of Substrate

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Abstract

A change in the admissible state space of a dynamical or combinatorial system need not be represented as destruction of structure. It may instead be represented as a change in the space of realizable transformations. This paper studies that distinction using finite token-configuration graphs with symmetry.

For a finite substrate graph X , we consider configuration spaces $X^{\{r\}}$ obtained by allowing a distinguished vertex m to carry increasing occupancy. The transition from $X^{\{r\}}$ to $X^{\{r+1\}}$ introduces a new relative layer

$$\mathcal{D}_r = V(X^{\{r+1\}}) \setminus V(X^{\{r\}}).$$

Rather than treating this layer as a set of states simply removed when the capacity is reduced, we study the relative cohomology

$$D_r = \ker \left(\left(R_G(X^{\{r+1\}}) \rightarrow R_G(X^{\{r\}}) \right) \right),$$

where R_G denotes the symmetry-invariant residual sector of the configuration-edge cochain space.

The principal result is that the relative sector admits a canonical cone-quotient description. Collapsing the entire lower-capacity configuration space to a single cone point converts the relative problem into the first cohomology of a finite quotient graph. After removing symmetry-flippable edge orbits, one obtains

$$\boxed{\dim D_r = \beta_1(\mathcal{Q}_r) = |E(\mathcal{Q}_r)| - |V(\mathcal{Q}_r)| + c(\mathcal{Q}_r).}$$

The component term $c(\mathcal{Q}_r)$ is essential: the relative sector is not generally represented by the connected-graph formula $E - V + 1$. For connected ordinary token substrates, we prove that the quotient is connected whenever the newly admitted layer contains a vacancy, while full saturation produces a second isolated component.

A second structural result shows that symmetry-flippable edge orbits become self-loops in the quotient and therefore cannot affect its component count. This closes the connectivity argument required for the cone formula.

Computational experiments over 55 capacity transitions spanning 11 substrates and distinguished sites, capacities $K=3,4,5$, and stabilizer sizes from 1 through 12 , found agreement with the structural formula in every tested case. In particular, all 55 restriction maps were surjective.

The resulting picture separates three phenomena that can otherwise appear as one: change in admissibility, loss of invariant visibility under symmetry, and genuine relative transformation structure. The final section discusses the implications for the broader constraint-first Nexus framework, while keeping those interpretations distinct from the mathematical results.

1. Introduction: What Does It Mean for a System to Change?

A system is commonly represented by its states and by rules describing transitions between those states. Under this representation, changing the system often appears as changing its list of states, adding or deleting transitions, or modifying the parameters governing them.

There is another possibility.

A system may change because its *space of admissible transformations changes*.

This distinction is important whenever one state space is embedded in another. If

$Y_o \subset Y$,

then the passage from Y_o to Y introduces states and transformations that cannot be represented inside Y_o . When one subsequently returns from Y to Y_o , those structures appear to have been lost.

But “not represented in the smaller space” is not equivalent to “destroyed.”

The natural mathematical question is therefore not

What disappeared?

but

****What transformation structure exists in the larger space relative to the smaller one?***

This paper develops that question for finite token-configuration graphs.

The construction is deliberately elementary at its foundation. A finite substrate graph X defines a configuration graph whose vertices are legal token configurations and whose edges represent one-token moves. A distinguished vertex m is allowed increasing occupancy. The resulting sequence

$X^{\{0\}}, X^{\{1\}}, X^{\{2\}}, \dots$

provides a controlled family of changes in admissible configuration space.

At each step,

$X^{\{r\}} \subset X^{\{r+1\}},$

and the newly admitted configurations form a layer

$\mathcal{D}_r$.

The central object of this paper is the relative sector associated with that layer.

The key observation is that the entire lower-capacity space can be collapsed to a single point. This does not delete the lower space from the relative problem. It records it as a boundary condition:

$$C \bullet (X^{\{r+1\}}, X^{\{r\}}) = \widetilde{C} \bullet \left(X^{\{r+1\}} / X^{\{r\}} \right).$$

The lower-capacity configuration space becomes a ****cone point****.

Once this is done, the relative problem becomes graph topology.

Symmetry introduces a second reduction. An oriented transformation that can be reversed by a symmetry operation cannot support a nonzero invariant antisymmetric cochain. Such a transformation becomes invisible in the invariant sector. In the quotient representation this is particularly simple: a flippable orbit becomes a self-loop.

Self-loops do not change connected components.

This observation turns what initially appeared to be a computational rank condition into a structural theorem.

The relative residual dimension is therefore governed by the number of symmetry-visible transformation edges, the number of symmetry-visible new states, and the number of connected components:

$$\boxed{\dim \mathcal{D}_r = E_Q - V_Q + c_Q.}$$

The result has two important consequences.

First, the commonly encountered expression

$$E_Q - V_Q + 1$$

is not the fundamental formula. It is only the connected case.

Second, a zero residual at the terminal layer does not generally mean that a quantity called “price” has been exhausted. At the terminal layer there is a single new configuration, and the residual dimension is instead controlled by the stabilizer's action on the neighbors of the distinguished site:

$$\boxed{D_{K-1} = \# \operatorname{bigl}(N(m) / \operatorname{operatorname{Stab}}(m) \operatorname{bigr}) - 1.}$$

Thus local symmetry and global saturation are distinct mechanisms.

The remainder of the paper establishes these statements formally, verifies them computationally, and then discusses their interpretation within the constraint-first Nexus framework.

2. Configuration Spaces and Capacity Layers

Let

$$X = (V, E)$$

be a finite connected graph.

Fix a distinguished vertex

$$m \in V.$$

We consider ‘K’-token configurations with exclusion away from ‘m’, while ‘m’ may carry additional occupancy.

For each ‘r’ ≥ 0, define ‘X^{r}’ as the configuration graph in which ‘m’ carries at least ‘r’ units of occupancy, with the remaining ‘K-r’ units subject to ordinary exclusion.

The precise configuration convention gives the nested sequence

$$X^{\{0\}} \subset X^{\{1\}} \subset X^{\{2\}} \subset \cdots$$

The newly admitted layer is

$$\boxed{\operatorname{mathcal{D}}_r = V(X^{\{r+1\}}) \setminus V(X^{\{r\}}).}$$

Every state in this layer has the form

$$m^{\{r+1\}} + U,$$

where

$$|U| = K - r - 1, \quad U \subseteq V \setminus \{m\}.$$

It is useful to introduce

$$k = K - r - 1.$$

Then the layer is parameterized by k -subsets of $V \setminus \{m\}$:

$$\boxed{\mathcal{D}_r = \{m^{r+1} + U \mid |U|=k\}.}$$

This reparameterization is important later because the combinatorial structure of the layer depends on the remaining occupancy k , while the distinguished multiplicity $r+1$ is fixed throughout the layer.

3. The Relative Sector

Let

$$Y = X^{(r+1)}, \quad Y_0 = X^{(r)}.$$

The relative cochain complex is

$$\boxed{C^\bullet(Y, Y_0).}$$

Equivalently, collapsing the entire subcomplex Y_0 to one point gives

$$\boxed{C^\bullet(Y, Y_0) = \widetilde{C^\bullet(Y/Y_0)}.}$$

Write the collapsed point as

$*$.

The important point is that Y_0 is not simply discarded. Its entire structure is represented by the single cone point.

Thus a transformation in the relative complex can be of two types:

1. an internal transformation

$$\mathcal{D}_r \xrightarrow{\quad} \mathcal{D}_r,$$

2. a cross transformation

$$\mathcal{D}_r \xrightarrow{\quad} *.$$

The latter records a transformation from the newly admitted layer into the previously admissible configuration space.

This is the geometric reason the cone point is essential.

4. Symmetry and Flippable Transformations

Let

$$G = \operatorname{Stab}(m)$$

be the symmetry group preserving the distinguished site.

The group acts on Y , Y_0 , and $\mathcal{D}_r$.

We consider the G -invariant relative cochains.

An oriented edge is written

$$e = [S \rightarrow T].$$

An invariant antisymmetric one-form satisfies

$$\omega(g \cdot e) = \omega(e)$$

and

$$\omega([T \rightarrow S]) = -\omega([S \rightarrow T]).$$

Suppose some $g \in G$ reverses the edge:

$$gS = T, \quad gT = S.$$

Then invariance gives

$$\omega(e) = \omega(g \cdot e) = \omega(-e) = -\omega(e),$$

and therefore

$$\boxed{\omega(e) = 0.}$$

Such an edge orbit is called **flippable**.

Consequently:

a flippable orbit contributes no invariant one-form degree of freedom;

a non-flippable orbit contributes one scalar degree of freedom.

This motivates deleting flippable edge orbits from the quotient representation.

5. The Cone Quotient Graph

Define $\widehat{\mathcal{Q}}_r$ to be the quotient graph whose vertices are

$G \backslash \{\text{orbits of } \mathcal{D}_r\}$

together with the cone point

$*$.

Its edges are G -orbits of transformations touching $\mathcal{D}_r$, including both internal and cross transformations.

Now delete the flippable edge orbits.

The resulting graph is

$$\boxed{\mathcal{Q}_r}$$

Thus $\mathcal{Q}_r$ contains exactly the symmetry-visible transformation degrees of freedom of the relative layer.

Let

$$V_Q = |V(\mathcal{Q}_r)|, \quad E_Q = |E(\mathcal{Q}_r)|, \quad c_Q = c(\mathcal{Q}_r).$$

6. Cone-Quotient Theorem

We can now state the central structural result.

Theorem 1 — Relative invariant dimension is a quotient Betti number

Let $Y = X^{\{r+1\}}$, $Y_0 = X^{\{r\}}$, and let $G = \operatorname{Stab}(m)$. Let $\mathcal{Q}_r$ be the symmetry quotient of the relative layer after deletion of flippable edge orbits.

Then

$$\boxed{\dim C^1_G(Y, Y_0) = E_Q}$$

and

$$\boxed{\dim C^0_G(Y, Y_0) = V_Q - 1.}$$

Moreover,

$$\boxed{\dim \ker d_{\{0, G\}^{\{\mathrm{rel}\}}} = c_Q - 1.}$$

Therefore

$$\boxed{\operatorname{rank} d_{\{0, G\}^{\{\mathrm{rel}\}}} = V_Q - c_Q.}$$

Since the relative graph has no nontrivial C^2 contribution,

$$\boxed{\dim H^1_G(Y, Y_0) = E_Q - V_Q + c_Q.}$$

Hence

$$\boxed{\dim D_r = \dim H^1_G(Y, Y_0) = \beta_1(\mathcal{Q}_r).}$$

Proof

Each non-flippable edge orbit supports one independent invariant antisymmetric coefficient, while each flippable orbit is forced to zero. Therefore

$$\dim C^1_G(Y, Y_0) = E_Q.$$

Relative zero-cochains are functions on the G -orbits of new-layer vertices. The cone point is fixed to zero in the relative complex, giving

$$\dim C^0_G(Y, Y_0) = V_Q - 1.$$

A zero-cochain lies in the kernel of d_0 precisely when it is constant on every connected component. The component containing the cone point must have value zero. Hence

$$\dim \ker d_0 = c_Q - 1.$$

Therefore

$$\operatorname{rank} d_0 = (V_Q - 1) - (c_Q - 1) = V_Q - c_Q.$$

Subtracting the rank from the dimension of C^1_G gives

$$\dim H^1_G = E_Q - (V_Q - c_Q) = E_Q - V_Q + c_Q.$$

This is the first Betti number of the quotient graph. ■

7. Why the Component Term Matters

It is tempting to replace

$$E_Q - V_Q + c_Q$$

with

$$E_Q - V_Q + 1.$$

That replacement is valid only when

$$c_Q = 1.$$

The distinction is not cosmetic.

Consider the saturated example

$$C_3, \text{ where } K=4, \text{ where } r=1.$$

The new layer consists of a single configuration

$$m^2ab.$$

There is no vacancy, and therefore no transformation from this state back to the lower-capacity configuration space.

Consequently,

$$V_Q = 2, \text{ where } E_Q = 0, \text{ where } c_Q = 2.$$

Thus

$$\beta_1(\mathcal{Q}_r) = 0 - 2 + 2 = 0.$$

Without the cone-component term one would obtain

$$0 - 2 + 1 = -1,$$

which is impossible for a Betti number.

The cone therefore does not provide an ad hoc correction. The component term is the ordinary graph-theoretic term in

$$\boxed{\beta_1 = E - V + c.}$$

8. Connectivity of the Relative Layer

The next issue is determining when c_Q can exceed one.

Before flippable deletion, consider

$$\widehat{\mathcal{Q}}_r.$$

Suppose a new-layer configuration is

$$m^{r+1} + U$$

and there exists a vacancy

$$v \in V \setminus (\{m\} \cup U).$$

If

$$N(m) \not\subseteq U,$$

the configuration has a direct cross transformation to the cone point.

The interesting case is

$$N(m) \subseteq U.$$

There is then no direct cross edge.

However, because X is connected and there is a vacancy, choose a path from the vacancy toward $N(m)$. Along that path the vacancy can be moved through occupied sites.

The moves remain within $\mathcal{D}_r$, because none of them changes the occupancy of m .

Eventually a neighbor $v_j \in N(m)$ becomes vacant.

The configuration then admits the cross move

$$m \rightarrow v_j,$$

which reaches Y_o , and therefore reaches the cone point.

Hence:

$$\boxed{\text{non-saturated layer} \rightarrow \widehat{\mathcal{Q}}_r \text{ is connected to } \{*\}.}$$

9. Flippable Orbits Are Loops

The remaining question is whether deleting flippable edge orbits can disconnect the quotient.

It cannot.

Lemma 2 — Flippable edge orbits are quotient self-loops

If an edge

$$e = \{S, T\}$$

is flippable, there exists $g \in G$ such that

$$gS = T, \text{ and } gT = S.$$

Therefore

$$[S]_G = [T]_G.$$

Thus the edge orbit has identical endpoints in the quotient:

$$[S]_G \xrightarrow{\text{long}} [S]_G.$$

It is a self-loop.

Deleting self-loops cannot alter the connected components of a graph.

Therefore

$$\boxed{c(\mathcal{Q}_r) = c(\widehat{\mathcal{Q}}_r).}$$

Combining this with vacancy propagation gives:

$$\boxed{\text{non-saturated} \xrightarrow{\text{long}} c(\mathcal{Q}_r) = 1.}$$

10. Saturation and the Two Mechanisms

Full saturation occurs when every site other than m is occupied:

$$U = V \setminus \{m\}.$$

Since

$$|U| = K - r - 1,$$

full saturation is equivalent to

$$K - r - 1 = |V| - 1,$$

or

$$\boxed{K = |V| + r.}$$

At full saturation there is exactly one new-layer configuration.

There is no vacancy and therefore no cross transformation to the cone.

The quotient consists of two isolated vertices:

$$\mathcal{Q}_r = \{*\} \sqcup \mathcal{D}_r.$$

Thus

$$V_Q=2, \quad E_Q=0, \quad c_Q=2,$$

and

$$\boxed{D_r=0.}$$

Therefore, for connected ordinary token substrates with nonempty layers,

$$\boxed{c(\mathcal{Q}_r) > 1 \iff K = |V| + r.}$$

Equivalently,

$$\boxed{\begin{cases} c_Q=1, & K < |V| + r, \\ c_Q=2, & K = |V| + r. \end{cases}}$$

This separates two mechanisms that can otherwise be confused:

$$\boxed{\text{local symmetry} \neq \text{global saturation}.}$$

Local symmetry controls whether invariant transformation channels survive.

Global saturation controls whether the newly admitted layer is connected to the old configuration space at all.

11. The Terminal Layer

At

$$r=K-1,$$

the new layer contains the single configuration

$$m^K.$$

Assuming ' $K < |V|$ ', the layer is not globally saturated.

The only possible cross transformations are moves

$$m \rightarrow v, \quad v \in N(m).$$

Under the stabilizer ' $G = \operatorname{Stab}(m)$ ', these neighbors split into orbits.

Therefore

$$V_Q=2$$

and

$$E_Q = \# \backslash \operatorname{bigl}(N(m)/G \backslash \operatorname{bigr}).$$

The quotient is connected, so

$$\boxed{D_{K-1} = \# \backslash \operatorname{bigl}(N(m)/\operatorname{Stab}(m) \backslash \operatorname{bigr}) - 1.}$$

This gives an exact interpretation of the terminal layer.

The terminal residual vanishes if and only if the stabilizer is transitive on the neighbors of m :

$$\boxed{D_{K-1} = 0 \text{ iff } \operatorname{Stab}(m) \text{ is transitive on } N(m).}$$

Thus a terminal zero is not evidence that a transformation “price” has been exhausted.

It can instead be a consequence of local symmetry.

This distinction is essential.

12. The Capacity Defect

Define the restriction map

$$\sigma_r: R_G(X^{(r+1)}) \rightarrow R_G(X^{(r)}).$$

The capacity defect is

$$\boxed{D_r = \ker \sigma_r.}$$

The relative-cohomology identification gives

$$\boxed{D_r \cong H^1_G(X^{(r+1)}, X^{(r)}).}$$

Thus the downward map and the upward relative sector describe the same structure from opposite directions:

$$\boxed{R_G(X^{(r+1)}) \twoheadrightarrow R_G(X^{(r)})}$$

shows the apparent loss under restriction, while

$$\boxed{H^1_G(X^{(r+1)}, X^{(r)})}$$

exposes the transformation structure associated with that loss.

This yields a useful three-level distinction:

$$\boxed{\text{admissibility} \rightarrow \text{field-relative structure} \rightarrow \text{observer-visible structure}.}$$

In ordinary cohomology,

$$D_r^{\mathrm{ord}} = H^1(X^{(r+1)}, X^{(r)}),$$

while symmetry selects

$$D_r = H^1_1 G(X^{\{r+1\}}, X^{\{r\}}).$$

The difference

$\dim D_r^{\{\mathrm{ord}\}} - \dim D_r$

measures mathematical symmetry filtering.

It should not, by itself, be identified with physical information loss.

13. Computational Verification

The structural derivation was tested against a computational sweep of

$\boxed{55/55}$

capacity transitions.

The sweep covered:

11 substrate/site cases;

$K=3,4,5$;

stabilizer sizes ranging from 1 to 12 ;

multiple substrate families and distinguished-site choices.

For every tested transition,

$\boxed{\operatorname{coker} \sigma_r = 0.}$

The cone-quotient Betti calculation agreed with the directly computed invariant relative dimension in every case.

Thus the computational result is:

$\boxed{55/55 \text{ agreement}.}$

The significance of this result is methodological.

The 55 experiments are **verification of a derived formula**, not the basis from which the formula is inferred.

The logical order is:

$\boxed{\text{relative complex} \rightarrow \text{cone quotient} \rightarrow \text{invariant edge counting} \rightarrow \beta_1 \rightarrow \text{computation as verification}.}$

This distinction prevents the computational sweep from being mistaken for a proof by induction from examples.

14. What the Sequence ' p_r ' Actually Measures

Define

$$\boxed{p_r = \dim D_r = \beta_1(\mathcal{Q}_r).}$$

The sequence

$$p_0, p_1, p_2, \dots$$

is therefore a graded sequence of first Betti numbers associated with successive capacity layers.

Writing

$$k = K - r - 1,$$

the layer is

$$m^{r+1} + U, \quad |U| = k.$$

Hence one may equivalently define

$$P_X(k) = \beta_1(\mathcal{Q}(X, m, k))$$

and obtain

$$\boxed{p_r = P_X(K - r - 1).}$$

This changes the interpretation of the capacity ladder.

The sequence is not merely a list of residual dimensions.

It is a sequence describing how the topology of the symmetry-visible transformation quotient changes as occupancy is redistributed.

The next mathematical problem is therefore not another isolated falsification test, but the determination of the recurrence or closed form governing

$$\boxed{P_X(k).}$$

15. Change of Substrate

The capacity construction is one example of a broader principle.

Suppose the transformation field associated with a substrate ' X ' is denoted

$$\mathcal{T}_X.$$

A change of substrate

$$X \rightarrow X'$$

changes the admissible transformation field

$$\mathcal{T}_X \rightarrow \mathcal{T}_{X'}.$$

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This can occur through several mathematically distinct operations.

15.1 Extension

Adding transformations gives

$$\mathcal{T}_X \subseteq \mathcal{T}_{X'}$$

The larger space contains the original transformations and additional ones.

15.2 Re-factorisation

Subdivision can replace

$$T$$

by

$$T_2 T_1$$

The transformation has not necessarily been destroyed. Its realization has been refactorized into intermediate steps.

15.3 Identification

Contraction can identify states or transformation endpoints.

This operation is different from extension and subdivision because distinct states may become one state.

15.4 Symmetry enlargement

Increasing the symmetry group does not necessarily destroy an underlying transformation.

Instead it can remove its visibility from the invariant sector.

This motivates the compact distinction:

$\boxed{\text{extension adds realizations;}}$

$\boxed{\text{re-factorisation changes realization path;}}$

$\boxed{\text{identification merges realizations;}}$

$\boxed{\text{symmetry filters visibility.}}$

These should not be conflated.

16. Interpretation Within the Nexus Framework

The mathematical construction was developed as part of a broader constraint-first framework beginning with two primitives.

Co — distinction exists

A system must contain distinguishable states for information, observation, or structure to exist.

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C₁ — lawful continuation

Every distinction admits lawful continuation.

The primitive object is consequently not a static object but a transition relation.

A state is a temporary section through an ongoing transformation field.

In this language, the configuration graph provides an explicit finite realization of the transition-field idea:

$$\boxed{G=(V,E)}$$

with vertices representing distinguishable states and edges representing lawful transformations.

The present construction adds a further distinction.

When

$Y_o \subset Y$,

the larger system contains transformations that have no realization inside ' Y_o '.

The relative complex records precisely that difference.

Thus:

$$\boxed{\text{change of admissibility} \rightarrow \text{relative transformation sector}.}$$

The cone construction then shows that the lower state space need not be treated as annihilated. It can instead be compressed to a boundary point while retaining its role in the transformation topology.

This motivates the interpretation:

****Change of substrate does not require destruction of transformation structure. It changes admissible realization space; the relative sector records transformations present in the larger space but absent from the smaller, while symmetry determines which remain observable as invariant structure.****

This is an interpretation of the mathematical construction, not a replacement for it.

17. Transformation and Memory

The broader Nexus framework treats persistent structure as transformed history.

Under this interpretation, a configuration is not merely a point in an abstract state space. It is a present realization carrying the consequences of prior transformations.

The transition

$S \rightarrow S'$

changes the present configuration.

If the resulting configuration persists, that transformation is represented in the new state.

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In this sense,

$$\boxed{\text{motion transforms configuration}}$$

while

$$\boxed{\text{configuration constrains subsequent motion}}.$$

The present paper does not claim that this interpretation establishes a physical theory of matter or memory. Rather, it provides a finite mathematical setting in which one can study a precise version of the underlying idea:

A change in the admissible transformation field can leave a persistent structural residue in the new configuration space.

The relative cohomology

$$H^1_G(Y, Y_0)$$

is one mathematically explicit realization of such a residue.

Whether that residue should ultimately correspond to a physical quantity is a separate empirical question.

18. What Has Been Established

The results of this paper separate naturally into mathematical theorems, computational verification, and interpretation.

Established mathematically

$$\boxed{D_r \cong H^1_G(X^{(r+1)}, X^{(r)})}$$

under the stated configuration and symmetry construction.

The cone representation gives

$$\boxed{C^\bullet(Y, Y_0) = \widetilde{C^\bullet(Y/Y_0)}}.$$

The invariant relative complex reduces to the quotient graph:

$$\boxed{\dim D_r = \beta_1(\mathcal{Q}_r)}.$$

Flippable edge orbits are quotient self-loops:

$$\boxed{c(\mathcal{Q}_r) = c(\widehat{\mathcal{Q}_r})}.$$

For connected ordinary token substrates, non-saturated layers are connected to the cone point, while full saturation gives a second component.

The terminal-layer formula is

$$\boxed{D_{K-1} = \#(N(m)/\operatorname{Stab}(m)) - 1}.$$

Computationally verified

The 55-transition sweep produced

$$\boxed{55/55}$$

agreement, with

$$\boxed{\operatorname{coker}\sigma_r=0}$$

throughout the tested family.

Not established by this paper

The following interpretations remain hypotheses rather than theorems:

that 'D_r' is physical information loss;

that 'D_r' represents matter;

that the residual dimension is a physical mass or inertia;

that the graph construction directly models fundamental spacetime;

that a specific physical constant follows from the residual sequence.

These questions require independent physical definitions and empirical consequences.

19. Discussion

The central result can be summarized in one sentence:

$$\boxed{\text{A change in admissible transformation space has a relative topology.}}$$

The relative topology is not obtained by simply counting new states.

It depends on transformations between those states.

A layer may contain many new configurations while producing no new invariant cycle.

Conversely, a relatively small layer can contain an independent cycle.

This is why

$$\text{new states} \neq \text{new transformation channels}.$$

The cone construction makes the distinction explicit.

The old configuration space is represented by a single point, while the new layer retains its internal transformation structure and its connections back to the old space.

The first Betti number then counts independent symmetry-visible loops in that relative transformation structure.

This also clarifies the role of symmetry.

A transformation may exist in the underlying graph while disappearing from the invariant sector. It has not been removed from the transformation field. It has become indistinguishable under the chosen symmetry action.

Thus three operations must remain distinct:

$$\boxed{\text{removal} \neq \text{identification} \neq \text{invisibility}.}$$

The present construction is particularly useful because those distinctions can be represented algebraically rather than only verbally.

20. Conclusion

We began with a simple question:

What structure is created when a system's admissible transformation space changes?

For finite symmetric token configuration graphs, the answer can be stated precisely.

Given

$$Y_0 = X^{\{r\}} \subset Y = X^{\{r+1\}},$$

the newly admitted transformation sector is described by the relative cohomology

$$\boxed{H^1_G(Y, Y_0).}$$

Collapsing the lower configuration space to a cone point produces a finite quotient graph $\mathcal{Q}_r$. Symmetry-flippable transformations become self-loops and therefore do not affect connectivity. The resulting residual dimension is exactly

$$\boxed{\dim H^1_G(Y, Y_0) = \beta_1(\mathcal{Q}_r) = E_Q - V_Q + c_Q.}$$

The component term is essential.

For connected ordinary token substrates, a non-saturated layer is connected to the old configuration space through vacancy propagation. Only full saturation creates an additional isolated component.

The terminal layer provides an independent local result:

$$\boxed{D_{\{K-1\}} = \#(N(m)/\operatorname{Stab}(m)) - 1.}$$

Therefore terminal vanishing is governed by local symmetry rather than by a generic exhaustion of a transformation quantity.

The computational program, spanning 55 capacity transitions, agrees with the structural result in every tested case.

The larger significance is methodological.

Rather than beginning with a proposed physical quantity and attempting to interpret its numerical behavior, the construction begins with distinctions, admissible transformations, symmetry, and relative topology. Only after the mathematical structure is established is interpretation introduced.

The resulting hierarchy is:

$$\boxed{\text{distinction} \rightarrow \text{continuation} \rightarrow \text{admissibility} \rightarrow \text{relative transformation sector} \rightarrow \text{symmetry quotient} \rightarrow \text{topological residual}.}$$

Whether this hierarchy describes only a useful class of combinatorial systems or points toward a more general physical principle remains an open question.

That question is now sufficiently separated from the mathematics to be tested.

Appendix A — Notation

$X=(V,E)$

finite substrate graph.

$m \in V$

distinguished site.

$X^{\{r\}}$

capacity-`r` configuration graph.

$$Y=X^{\{r+1\}}, \quad Y_0=X^{\{r\}}.$$

$\mathcal{D}_r=V(Y) \setminus V(Y_0).$

$$G=\operatorname{Stab}(m).$$

$$\mathcal{D}_r=\ker \sigma_r.$$

$\mathcal{Q}_r$

symmetry-reduced relative quotient after deletion of flippable edge orbits.

$$E_Q, V_Q, c_Q$$

edge, vertex, and component counts of $\mathcal{Q}_r$.

$$\beta_1(\mathcal{Q}_r) = E_Q - V_Q + c_Q.$$

Appendix B — Computational Scope

The verification program covered 55 capacity transitions across 11 substrate/site configurations,
with

$$K=3,4,5$$

and stabilizer orders ranging from the trivial group to order '12'.

For every tested transition:

$\sigma_r=0$.

The directly calculated relative invariant dimensions agreed with the cone-quotient prediction in all
tested cases.

The purpose of the computational sweep was verification of the derived structural result, not
induction of the result from finite examples.

Appendix C — Locked Research Status

Current spine: Change $\rightarrow$ Layer $D_r \rightarrow$ Relative $C^{\bullet}(Y, Y_o) = \tilde{C}^{\bullet}(Y/Y_o) \rightarrow$ Cone $\ast \rightarrow$ Symmetry $G = \text{Stab}(m) \rightarrow$ Quotient $\hat{Q}_r \rightarrow$ Delete flippable loops $Q_r \rightarrow$ Cycle $\beta_1(Q_r) = E_Q - V_Q + c_Q \rightarrow p_r$.

Proved: $D_r \cong H^1_G(Y, Y_o)$; the cone-quotient Betti formula; flippable = quotient self-loop; vacancy propagation; the component result for connected ordinary token substrates; full-saturation characterization; and the terminal-layer formula.

Verified, not proved from examples: The 55/55 principal capacity sweep, the broader 492-transition check for the component and quotient claims, and the 47-case terminal-layer check.

Next object: $p_r = \beta_1(Q_r) = P_X(K - r - 1)$. The open mathematical problem is the recurrence or closed form governing $P_X(k)$ as k changes.

Appendix D — Broader Verification Scope

The broader 492-transition search is recorded here as an independent computational check, not as evidence from which the structural theorem is inferred. It found no nonempty-old-arena case with $c_Q > 1$, no mismatch between $\dim D_r$ and $\beta_1(Q_r)$, and no flippable orbit that failed to become a quotient self-loop.

The top-layer formula was independently checked in 47 nondegenerate cases with 47/47 agreement, including nonzero terminal defects for asymmetric distinguished sites.